

Research Notes on Differential privacy machine learning

Differential privacy machine learning

>> Section 1

Publications Calibrating noise to sensitivity in private data analysis, Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. 2006. In Proceedings of the Third conference on Theory of Cryptography (TCC'06). Springer-Verlag, Berlin, Heidelberg, 265–284. Calibrating Noise to Sensitivity in Private Data Analysis (This is the original publication of Differential Privacy, and not the eponymous article by Dwork that was published the same year.) Differential Privacy: A Survey of Results by Cynthia Dwork, Microsoft Research, April 2008 (Presents what was discovered during the first two years of research on differential privacy.) Differential Privacy: A Primer for a Non-Technical Audience, Alexandra Wood, Micah Altman, Aaron Bembek, Mark Bun, Marco Gaboardi, et al, Vanderbilt Journal of Entertainment & Technology Law/Vanderbilt Journal of Entertainment, Volume 21, Issue 1, Fall 2018. (A good introductory document, but definitely *not* for non-technical audiences!) Technology Factsheet: Differential Privacy by Raina Gandhi and Amritha Jayanti, Belfer Center for Science and International Affairs, Fall 2020 Differential Privacy and the 2020 US Census, MIT Case Studies in Social and Ethical Responsibilities of Computing, no. Winter 2022 (January). Differential Privacy and the 2020 US Census. Garfinkel, Simson (2025). Differential Privacy. MIT Press Essential Knowledge. MIT Press. doi:10.7551/mitpress/15354.001.0001. ISBN 9780262551656. Bowen, Claire McKay and Simson Garfinkel, The Philosophy of Differential Privacy, AMS Notices, November 2021.

>> Section 2

A $\{ \text{displaystyle } A \}$ and B $\{ \text{displaystyle } B \}$ (where A $\{ \text{displaystyle } A \}$ contains the group and B $\{ \text{displaystyle } B \}$ does not). In this case M $\blacksquare$ T $\{ \text{displaystyle } M \circlearrowright C \}$ is $(\epsilon \times c \times h)$ $\{ \text{displaystyle } (\text{wrepsilon } \times c \times h) \}$ -differentially private.

>> Section 3

Public purpose considerations There are several public purpose considerations regarding differential privacy that are important to consider, especially for policymakers and policy-focused audiences interested in the social opportunities and risks of the technology:

>> Section 4

Group privacy In general, ϵ -differential privacy is designed to protect the privacy between neighboring databases which differ only in one row. This means that no adversary with arbitrary auxiliary information can know if one particular participant submitted their information. However this is also extendable. We may want to protect databases

differing in c items: $\Pr[A(D_1) \in S] \leq \exp(\epsilon c) \cdot \Pr[A(D_2) \in S]$. Thus setting ϵ instead to ϵ / c achieves the desired result (protection of c items). In other words, instead of having each item ϵ -differentially private protected, now every group of c items is ϵ -differentially private protected (and each item is (ϵ / c) -differentially private protected).

>> Section 5

Data utility and accuracy. The main concern with differential privacy is the trade-off between data utility and individual privacy. If the privacy loss parameter is set to favor utility, the privacy benefits are lowered (less “noise” is injected into the system); if the privacy loss parameter is set to favor heavy privacy, the accuracy and utility of the dataset are lowered (more “noise” is injected into the system). It is important for policymakers to consider the trade-offs posed by differential privacy in order to help set appropriate best practices and standards around the use of this privacy preserving practice, especially considering the diversity in organizational use cases. It is worth noting, though, that decreased accuracy and utility is a common issue among all statistical disclosure limitation methods and is not unique to differential privacy. What is unique, however, is how policymakers, researchers, and implementers can consider mitigating against the risks presented through this trade-off. Data privacy and security. Differential privacy provides a quantified measure of privacy loss and an upper bound and allows curators to choose the explicit trade-off between privacy and accuracy. It is robust to still unknown privacy attacks. However, it encourages greater data sharing, which if done poorly, increases privacy risk. Differential privacy implies that privacy is protected, but this depends very much on the privacy loss parameter chosen and may instead lead to a false sense of security. Finally, though it is robust against unforeseen future privacy attacks, a countermeasure may be devised that we cannot predict.

>> Section 6

Differential privacy (DP) is a mathematically rigorous framework for releasing statistical information about

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datasets while protecting the privacy of individual data subjects. It enables a data holder to share aggregate patterns of the group while limiting information that is leaked about specific individuals. This is done by injecting carefully calibrated noise into statistical computations such that the utility of the statistic is preserved while provably limiting what can be inferred about any individual in the dataset. Another way to describe differential privacy is as a constraint on the algorithms used to publish aggregate information about a statistical database which limits the disclosure of private information of records in the database. For example, differentially private algorithms are used by some government agencies to publish demographic information or other statistical aggregates while ensuring confidentiality of survey responses, and by companies to collect information about user behavior while controlling what is visible even to internal analysts. Roughly, an algorithm is differentially private if an observer seeing its output cannot tell whether a particular individual's information was used in the computation. Differential privacy is often discussed in the context of identifying individuals whose information may be in a database. Although it does not directly refer to identification and reidentification attacks, differentially private algorithms provably resist such attacks.

>> Section 7

Early research leading to differential privacy In 1977, Tore Dalenius formalized the mathematics of cell suppression. Tore Dalenius was a Swedish statistician who contributed to statistical privacy through his 1977 paper that revealed a key point about statistical databases, which was that databases should not reveal information about an individual that is not otherwise accessible. He also defined a typology for statistical disclosures. In 1979, Dorothy Denning, Peter J. Denning and Mayer D. Schwartz formalized the concept of a Tracker, an adversary that could learn the confidential contents of a statistical database by creating a series of targeted queries and remembering the results. This and future research showed that privacy properties in a database could only be preserved by considering each new query in light of (possibly all) previous queries. This line of work is sometimes called query privacy, with the final result being that tracking the impact of a query on the privacy of individuals in the database was NP-hard.